

LAB EXERCISE: SCHEMATICS WITH ESP32

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Part A

For each of these circuits:

1. Calculate the total resistance in ohms
2. Calculate the total current from the power supply

Series Circuit:

Supply voltage = 125 V

Resistors: 20Ω, 30Ω, 50Ω

It's a series circuit.

Step 1: Calculate total resistance

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 20 + 30 + 50$$

$$R_T = 100\Omega$$

Total resistance = 100 Ω.

Step 2: Calculate total current

Use Ohm's Law:

$$V = IR$$

$$I = \frac{V}{R}$$

$$V = 125V$$

$$R_T = 100\Omega$$

$$I_T = \frac{125}{100}$$

$$I_T = 1.25A$$

Parallel Circuit:

There are three resistors 20Ω, 100Ω, 50Ω

Step 1: Calculate total resistance

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\frac{1}{20} = 0.05$$

$$\frac{1}{100} = 0.01$$

$$\frac{1}{50} = 0.02$$

$$\frac{1}{R_T} = 0.05 + 0.01 + 0.02$$

$$\frac{1}{R_T} = 0.08$$

$$R_T = \frac{1}{0.08}$$

$$\boxed{R_T = 12.5\Omega}$$

Total resistance = 12.5 Ω .

Step 2: Calculate total current

$$I_T = \frac{V}{R_T}$$

$$V = 125V$$

$$R_T = 12.5\Omega$$

$$I_T = \frac{125}{12.5}$$

$$\boxed{I_T = 10A}$$

3. Calculate the current through each resistor

Series circuit:

In a series circuit, the current is the same through every resistor.

Therefore:

$$I_1 = I_2 = I_3 = 1.25A$$

Resistor	Current
20 Ω	1.25 A
30 Ω	1.25 A
50 Ω	1.25 A

Parallel circuit:

In parallel, use:

$$I = \frac{V}{R}$$

Since each resistor has 125 V across it:

20 Ω :

$$I_1 = \frac{125}{20} = \boxed{6.25A}$$

100 Ω :

$$I_2 = \frac{125}{100} = \boxed{1.25A}$$

50 Ω :

$$I_3 = \frac{125}{50} = \boxed{2.5A}$$

Resistor	Current
20 Ω	6.25 A
100 Ω	1.25 A
50 Ω	2.5 A

Check: 6.25+1.25+2.5=10A

4. Calculate the Voltage drop through each resistor

Use:

$$V = IR$$

Series circuit:

20 Ω resistor:

$$V_1 = 1.25 \times 20$$

$$\boxed{V_1 = 25V}$$

30 Ω resistor:

$$V_2 = 1.25 \times 30$$

$$\boxed{V_2 = 37.5V}$$

50 Ω resistor:

$$V_3 = 1.25 \times 50$$

$$\boxed{V_3 = 62.5V}$$

Check:

$$25 + 37.5 + 62.5 = 125V$$

Resistor	Voltage drop
20 Ω	25 V
30 Ω	37.5 V
50 Ω	62.5 V

Parallel circuit:

In a parallel circuit, the voltage across every branch is equal to the supply voltage.

$$V_1 = V_2 = V_3 = 125V$$

Resistor	Voltage drop
20 Ω	125 V
100 Ω	125 V
50 Ω	125 V

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

Series circuit:

$$P = V \times I$$

$$P_{20} = 25 \text{ V} \times 1.25 \text{ A} = 31.25 \text{ W}$$

$$P_{30} = 37.5 \text{ V} \times 1.25 \text{ A} = 46.875 \text{ W}$$

$$P_{50} = 62.5 \text{ V} \times 1.25 \text{ A} = 78.125 \text{ W}$$

Parallel circuit:

$$P = V \times I$$

$$P_{20} = 125 \text{ V} \times 6.25 \text{ A} = 781.25 \text{ W}$$

$$P_{100} = 125 \text{ V} \times 1.25 \text{ A} = 156.25 \text{ W}$$

$$P_{50} = 125 \text{ V} \times 2.5 \text{ A} = 312.5 \text{ W}$$